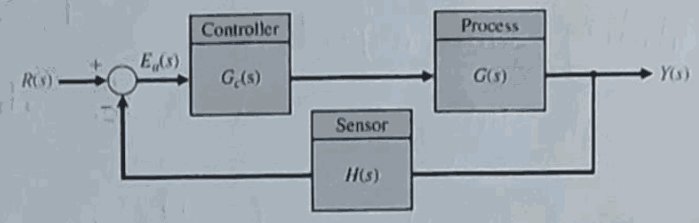
Question 3. (25%)

A feedback control system is shown below with the following transfer function definitions.



$$G_c(s) = K; \quad G(s) = \frac{s+40}{s(s+10)}; \quad H(s) = \frac{1}{s+20}$$

$$CLTF = F(s) = \frac{k\,G(s)}{1 + k\,G(s)\,H(s)}$$

$$= \frac{\dfrac{k(s+40)}{s(s+10)}}{1 + \dfrac{k(s+40)}{s(s+10)(s+20)}}$$

$$= \frac{\dfrac{k(s+40)}{s(s+10)}}{\dfrac{s(s+10)(s+20) + k(s+40)}{s(s+10)(s+20)}}$$

$$F(s) = \frac{k(s+40)(s+20)}{s(s+10)(s+20) + k(s+40)}$$

$$e_{ss} = \lim_{s\to 0} s\,E(s) = \lim_{s\to 0} \frac{s\,R(s)}{1 + F(s)}$$

For unit step $R(s) = \frac{1}{s}$

$$e_{ss} = \lim_{s\to 0} \frac{1}{1 + \underbrace{F(s)}_{K_p}}$$

$$k_p = \lim_{s\to 0} \frac{k(s+40)(s+20)}{s(s+10)(s+20) + k(s+40)} = \frac{k(40)(20)}{k(40)}$$

$e_{ss} = \frac{1}{21}$ For unit step input

For ramp input $R(s) = \frac{1}{s^2}$

$$e_{ss} = \lim_{s \to 0} \frac{1/s}{1 + F(s)} = \lim_{s \to 0} \frac{1}{s + sF(s)}$$

$$= \lim_{s \to 0} \frac{1}{sF(s)} = \frac{1}{k_a}$$

When $k_a = \lim_{s \to 0} sF(s) =$

$$\lim_{s \to 0} \frac{s\left(k\,(s+40)\,(s+20)\right.}{s(s+10)\,(s+20) + k(s+40)} = 0$$

$\Rightarrow e_{ss} = \infty$